

1 Package raddenest

1.1 Functions and variables for package raddenest

raddenest (*expr*) [Function]

Denests different classes of nested radical expressions appearing as subexpressions in *expr*. Most of the implemented algorithms are specific to square roots, but some handle higher roots.

raddenest is able to denest some or all of the denestable expressions of the following types:

1. square roots of sums of multiple unnested square roots;
2. $\sqrt{a + b\sqrt{r}}$ where a , b , and r are linear combinations of square roots of positive rationals;
3. expressions of the general form $\sqrt{a + b\sqrt{r}}$ are denested numerically (using square roots or fourth roots) or, in some cases, symbolically (assumptions allowing);
4. $(a + b\sqrt{r})^{\frac{1}{n}}$ with rational a , b , r (r positive) and integer n ;
5. $\sqrt{a^{\frac{1}{3}} + b^{\frac{1}{3}}}$ with rational a and b .

In particular, all expressions denested by **sqrtdenest** are also denested by **raddenest**.

Interpretation of radical expressions:

The results given by **raddenest** are consistent with Maxima's default interpretation for n -th roots of a real:

- $a^{\frac{1}{n}}$ with positive a corresponds to the positive real branch;
- $\sqrt{a}$ with negative a corresponds to $i\sqrt{-a}$;
- $a^{\frac{1}{3}}$ with negative a is interpreted using the real branch of the cube root if **domain=real**; with **domain=complex** subexpressions involving roots other than square roots are left unchanged.

Examples:

```
(%i1) load (raddenest)$
(%i2) sqrt(sqrt(16+2*sqrt(55-10*sqrt(29))-2*sqrt(29))-sqrt(5))$
(%i3) raddenest(%);

                                     1/4
(%o3)      (11 - 2 sqrt(29))
(%i4) sqrt(1+1/(7+4*sqrt(3)));

                                     1
(%o4)      sqrt(----- + 1)
                4 sqrt(3) + 7

(%i5) raddenest(%);
(%o5)      sqrt(6) - sqrt(2)
```

```

(%i6) sqrt(8*sqrt(2)+2*sqrt(5)-18);
                                7/2
(%o6)      sqrt(2 sqrt(5) + 2  - 18)
(%i7) raddenest(%);
(%o7)      ((- sqrt(10)) + sqrt(5) + sqrt(2) + 1) %i
(%i8) raddenest(sqrt(5^(1/3)-4^(1/3)));
                                1/3      1/3      1/3
                                (- 25  ) + 20  + 2
(%o8)      -----
                                3
(%i9) raddenest((41-29*sqrt(2))^(1/5));
(%o9)      1 - sqrt(2)
(%i10) assume(y>0)$
(%i11) raddenest(sqrt(9*y+6*x^2*sqrt(y)+x^4));
                                2
(%o11)      3 sqrt(y) + x
(%i12) a: (2-sqrt(5))^(1/3)$
(%i13) raddenest(a);
                                1      sqrt(5)
                                - - ----
                                2      2
(%i14) raddenest(a), domain:'complex;
                                1/3
(%o14)      (2 - sqrt(5))

```

Limitations:

raddenest may in some cases only lower the index of a root without actually decreasing the absolute nesting depth of *expr*:

```

(%i1) load (raddenest)$
(%i2) raddenest((5*sqrt(2)+7)^(1/6));
(%o2)      sqrt(sqrt(2) + 1)

```

References:

1. Allan Borodin, Ronald Fagin, John E. Hopcroft, and Martin Tompa: *Decreasing the Nesting Depth of Expressions Involving Square Roots* J. Symbolic Computation (1985) 1, 169-188
<http://researcher.watson.ibm.com/researcher/files/us-fagin/symb85.pdf>
2. David J. Jeffrey and Albert D. Rich: *Simplifying Square Roots of Square Roots by Denesting in Computer Algebra Systems: A Practical Guide* M.J. Wester, Ed., Wiley 1999
<http://www.cybertester.com/data/denest.pdf>
3. Thomas J. Osler: *Cardan Polynomials and the Reduction of Radicals* Mathematics Magazine 74(1), Feb. 2001
<http://www.rowan.edu/open/depts/math/osler/mathmag026-032.pdf>
4. Mascha Honsbeek: *Radical Extensions and Galois Groups*, Chapter 3 (PhD Thesis)

http://www.math.kun.nl/~bosma/students/honsbeek/M_Honsbeek_thesis.pdf

5. SymPy `sqrtdenest()` <http://www.sympy.org>

Appendix A Function and variable index

`raddenest` 1

(Index is nonexistent)